

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of the project is to create a system for the client, DriverPass, that assists students in preparing for their driving tests
- The client, DriverPass, is a company that wants to provide students with access to online practice exams and on-the-road training to better prepare them for driving tests.
- The system will allow customers to register, schedule driving lessons, take online tests, and track their progress.
- The system will allow employees to manage users, appointments, and system data.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass recognized that many people were failing their DMV driving tests due to lack of proper training.
- Their solution is to provide both online learning and in-person driving lessons.
- The system will support scheduling lessons, managing training packages, and tracking student progress.
- The system will connect to the DMV to receive updates, so the learning material stays current.
- The system will be web-based, making it accessible from anywhere, and will store all customer and business data in a secure fashion.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The system will allow users to create secure accounts to login.
- Customers will be able to schedule, cancel, and modify driving appointments.
- The system will track lesson details, including instructor, student, car, and time.
- Students should be able to do online practice tests and track scores, progress, and completion status.
- Employees should be allowed to manage users, reset passwords, and control access based on roles.
- The system should maintain accurate records for all changes and generate reports that can be downloaded.
- The system should receive updates from the DMV to keep all content current.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system should operate as a web-based application accessible through internet-enabled devices.
- The system should perform efficiently under high demand and high user engagement.
- The system should maintain fast response times even during peak usage.
- The system should support continuous availability since users depend on it for learning access.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system depends on internet access and compatible devices for all users.]
- The system may be limited by budget and available resources, affecting advanced features.
- The system should run on multiple

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system should distinguish users through secure login credentials and role-based access.
- The system should ensure accurate tracking of student progress and reporting.
- The system should maintain data accuracy for assignments, grades, and reports.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system should support scalability to handle growth in users and data over time.
- The system should allow admins to add, remove, or modify users without changing code.
- The system should adapt to platform and software updates with minimal disruption.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a “brute force” hacking attempt? What happens if the user forgets their password?

- The system should require secure authentication for login.
- The system should use data protection measures to safeguard information.
- The system should include regular data backups for recovery in case of failure.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with “The system shall . . .” For example, one functional requirement might be, “The system shall validate user credentials when logging in.”

- The system shall allow students, instructors, and admins to log in securely with role-based access.
- The system shall allow instructors to manage learning materials.
- The system shall allow students to submit assignments electronically.
- The system shall provide communication and collaboration tools.
- The system shall support testing and assessment materials.
- The system shall track student progress and generate reports for administrators.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The system should provide an intuitive and accessible interface for all users.
- The interface should support students, instructors, and administrators with different roles and permissions.
- The system should be easy to learn, especially with user training provided beforehand.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- All users have internet access and devices compatible with the LMS
- All users will receive training on the LMS before being required to use it independently

Limitations

*Any system you build will naturally have limitations. What limitations do you see in your system design?
What limitations do you have as far as resources, time, budget, or technology?*

- The company might have a limited budget and resources for advanced functions
- Some users may resist or not understand how to use the system effectively

Gantt Chart

DrivePass Gantt Chart



Gantt Legend
Requirement Phase
Design Phase
Development Phase
Customer Approval